Mutual Fund Analytics & Performance Dashboard

Bluestock Fintech Capstone Project

Submitted By

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Organization

Bluestock Fintech

Project Duration

June 2026 – July 2026

Technologies Used

- Python
- Pandas
- NumPy
- SQLite
- SQL
- Power BI
- Jupyter Notebook
- Git & GitHub

Submitted On

July 2026

Executive Summary

This project presents a complete Mutual Fund Analytics solution developed as part of the Bluestock Fintech Capstone Project. The objective was to build an end-to-end data analytics pipeline capable of processing mutual fund datasets, performing financial performance analysis, generating business insights, and visualizing results through an interactive Power BI dashboard.

The project begins with an automated ETL pipeline that ingests raw CSV datasets, validates and cleans the data, and loads the processed data into a SQLite database. Exploratory Data Analysis (EDA) was performed to study industry trends, mutual fund performance, investor behaviour, portfolio allocation, and benchmark movements.

Financial performance metrics including Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Tracking Error, Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), and Herfindahl-Hirschman Index (HHI) were calculated to evaluate the return and risk characteristics of different mutual fund schemes.

Advanced analytics such as Rolling Sharpe Ratio analysis, Investor Cohort Analysis, SIP Continuity Analysis, Sector Concentration Analysis, and a Risk-Based Mutual Fund Recommendation System were implemented to provide deeper insights for investors.

The final Power BI dashboard contains four interactive pages with slicers and visualizations that help analyse industry growth, compare fund performance, understand investor behaviour, and monitor SIP trends. The project demonstrates practical application of Python, SQL, SQLite, Power BI, and financial analytics techniques in solving real-world investment analysis problems.

Project Objectives

The primary objective of this project is to design and develop a complete Mutual Fund Analytics System capable of collecting, processing, analysing, and visualising mutual fund data to support data-driven investment decisions.

Objectives

- Build an automated ETL pipeline to ingest, clean, validate, and process mutual fund datasets.
- Store cleaned data efficiently in a SQLite database for analysis.
- Perform Exploratory Data Analysis (EDA) to identify trends in AUM, SIP inflows, investor behaviour, and portfolio holdings.
- Calculate key financial performance metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Tracking Error, VaR, and CVaR.
- Develop advanced analytics such as Investor Cohort Analysis, SIP Continuity Analysis, Rolling Sharpe Ratio, and Portfolio Concentration (HHI).
- Build a risk-based mutual fund recommendation system using historical performance and risk category.
- Design an interactive Power BI dashboard for visualising industry trends, fund performance, investor analytics, and SIP market statistics.
- Present actionable insights and recommendations based on the analysis.

Problem Statement

The Indian mutual fund industry has experienced significant growth in recent years, leading to an increasing number of investment schemes, investors, and financial products. While large volumes of data are generated daily, transforming this information into meaningful insights remains a challenge.

Investors often struggle to compare mutual funds based on both return and risk. Traditional comparisons based solely on historical returns ignore important risk indicators such as volatility, downside risk, drawdown, and benchmark performance. Similarly, fund houses require efficient tools to analyse investor behaviour, SIP continuity, portfolio diversification, and overall market trends.

This project addresses these challenges by developing an end-to-end Mutual Fund Analytics System that integrates data engineering, financial analytics, and business intelligence into a single workflow. The solution combines automated ETL processes, database management, statistical analysis, advanced risk metrics, and interactive dashboards to provide comprehensive insights into mutual fund performance and investor behaviour.

The final outcome enables users to evaluate mutual funds using multiple financial indicators rather than relying on returns alone, supporting more informed investment decisions.

Project Architecture

The Mutual Fund Analytics project follows a structured end-to-end data analytics pipeline. Data is collected from multiple raw datasets, cleaned using Python scripts, stored in a SQLite database, analysed using Jupyter Notebook, and finally visualised in an interactive Power BI dashboard.

Project Workflow

Raw CSV Files

↓

Python ETL Pipeline

(Data Cleaning & Validation)

↓

SQLite Database

↓

Jupyter Notebook

(EDA + Performance Analytics + Advanced Analytics)

↓

Power BI Dashboard

↓

Business Insights & Recommendations

Technology Stack

Technology	Purpose
Python	Data processing and analytics
Pandas	Data cleaning and manipulation
NumPy	Numerical computations
SQLite	Database storage
SQL	Data querying
Jupyter Notebook	Analysis and documentation
Power BI	Interactive dashboard development
Git & GitHub	Version control
VS Code	Python development
Microsoft Excel	Initial data inspection

The project integrates Python, SQL, SQLite, Power BI, and Jupyter Notebook to build a complete end-to-end analytics pipeline. Python scripts automate data ingestion and cleaning, SQLite stores structured datasets, Jupyter Notebook performs analytical computations, and Power BI provides interactive visualizations for business users.

Dataset Description

The project uses ten primary datasets covering mutual fund schemes, NAV history, investor transactions, benchmark indices, portfolio holdings, SIP inflows, and industry statistics.

Dataset	Description	Records
01_fund_master.csv	Mutual fund master details	40
02_nav_history.csv	Historical NAV prices	46,000
03_aum_by_fund_house.csv	Assets under management	90
04_monthly_sip_inflows.csv	Monthly SIP statistics	48
05_category_inflows.csv	Category-wise inflows	144
06_industry_folio_count.csv	Industry folio statistics	21
07_scheme_performance.csv	Fund performance metrics	40
08_investor_transactions.csv	Investor transactions	32,778
09_portfolio_holdings.csv	Portfolio holdings	322
10_benchmark_indices.csv	Benchmark index history	8,050

These datasets collectively provide information required for financial performance evaluation, investor behaviour analysis, portfolio concentration analysis, benchmark comparison, and dashboard reporting.

ETL (Extract, Transform, Load) Pipeline

The ETL (Extract, Transform, Load) pipeline was developed to automate the complete data preparation process for the Mutual Fund Analytics project. The pipeline reads raw datasets, validates the data, performs cleaning and transformation operations, and loads the processed datasets into a SQLite database. Automating these steps ensures consistency, reduces manual effort, and prepares reliable data for further analysis.

ETL Workflow

1. Extract raw mutual fund datasets.
2. Validate dataset structure and data types.
3. Clean missing values, duplicates, and invalid records.
4. Standardize date formats and numerical fields.
5. Save cleaned datasets to the processed folder.
6. Load processed datasets into SQLite.
7. Verify database tables and row counts.

Scripts Developed

Script	Purpose
data_ingestion.py	Reads raw datasets
clean_nav_history.py	Cleans NAV data
clean_scheme_performance.py	Cleans performance dataset
clean_investor_transactions.py	Cleans transaction dataset
load_to_sqlite.py	Loads data into SQLite
verify_sqlite_data.py	Verifies database
etl_pipeline.py	Runs the complete ETL process
live_nav_fetch.py	Fetches latest NAV from MF API

Figure 1: ETL Pipeline – Data Ingestion

```
Running: data_ingestion.py
============================================================
BLUESTOCK MUTUAL FUND DATA INGESTION
============================================================

Found 16 CSV files.

============================================================
Reading: 01_fund_master.csv

Shape: (40, 15)

Data Types:
amfi_code                int64
fund_house                 str
scheme_name                str
category                   str
sub_category               str
plan                       str
launch_date                str
benchmark                  str
expense_ratio_pct      float64
exit_load_pct          float64
min_sip_amount           int64
min_lumpsum_amount       int64
fund_manager               str
risk_category              str
sebi_category_code         str
dtype: object
```

```
Reading: 02_nav_history.csv

Shape: (46000, 3)

Data Types:
amfi_code      int64
date             str
nav          float64
dtype: object

First 5 Rows:
   amfi_code        date      nav
0     119551  2022-01-03  54.3856
1     119551  2022-01-04  54.3474
2     119551  2022-01-05  54.6869
3     119551  2022-01-06  55.4550
4     119551  2022-01-07  55.3692
============================================================
Reading: 03_aum_by_fund_house.csv

Shape: (90, 5)

Data Types:
date                   str
fund_house             str
aum_lakh_crore     float64
aum_crore            int64
num_schemes          int64
dtype: object
```

Figure 2: SQLite Database Loading

```
Running: load_to_sqlite.py
==================================================
CONNECTED SUCCESSFULLY TO SQLITE DATABASE
==================================================
dim_fund loaded
fact_nav loaded
fact_transactions loaded
fact_performance loaded
fact_aum loaded
monthly_sip_inflows loaded
category_inflows loaded
industry_folio_count loaded
portfolio_holdings loaded
benchmark_indices loaded

All datasets loaded successfully!
load_to_sqlite.py completed successfully.
```

Figure 3: ETL Pipeline Completed Successfully

```
----------------------------------------
verify_sqlite_data.py completed successfully.

============================================================
ETL Pipeline Completed Successfully!
============================================================
PS C:\Users\Pranav Shinde\Documents\Bluestock_MF_Capstone>
```

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure, quality, and characteristics of the mutual fund datasets before conducting performance analysis. Various visualizations were created to identify trends, distributions, relationships, and potential insights related to mutual funds, investors, and market behaviour.

The analysis covers industry growth, investor demographics, SIP patterns, sector allocation, portfolio diversification, and benchmark movements. These insights helped validate the data and provided a foundation for the financial analytics performed in later stages of the project.

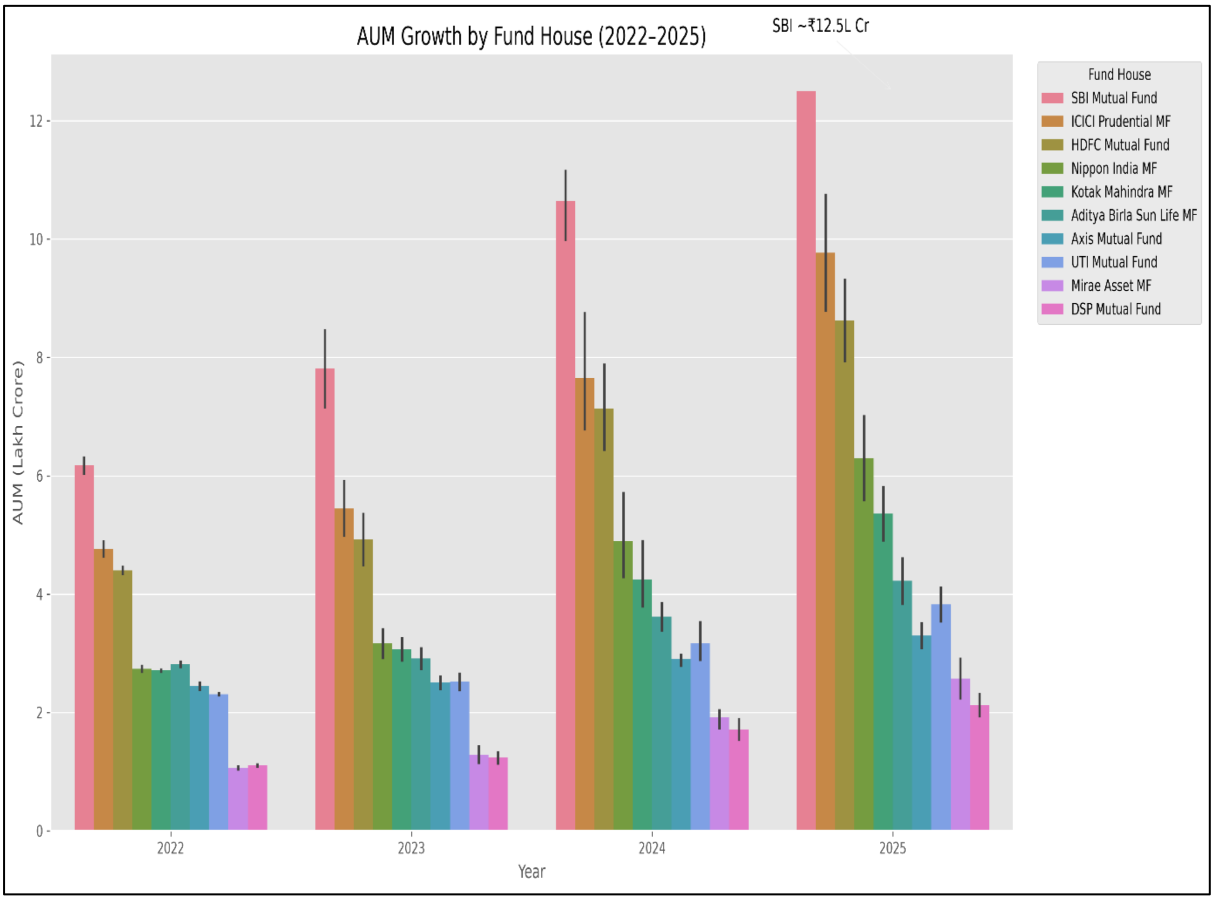


Figure 4.1 – Growth in Assets Under Management (AUM)

The chart compares the Assets Under Management (AUM) of different mutual fund houses from 2022 to 2025. SBI Mutual Fund consistently maintained the largest AUM, indicating its strong market position and investor confidence.

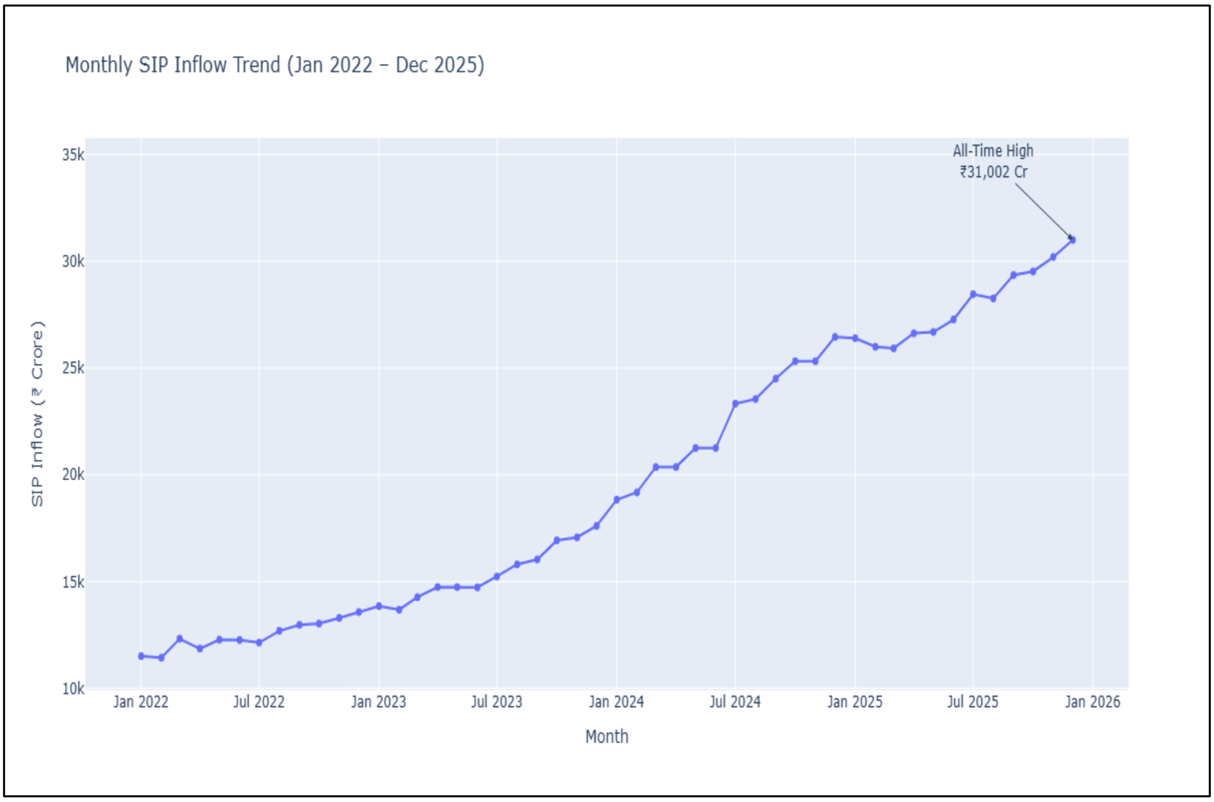


Figure 4.2 – Monthly SIP Inflow Trend

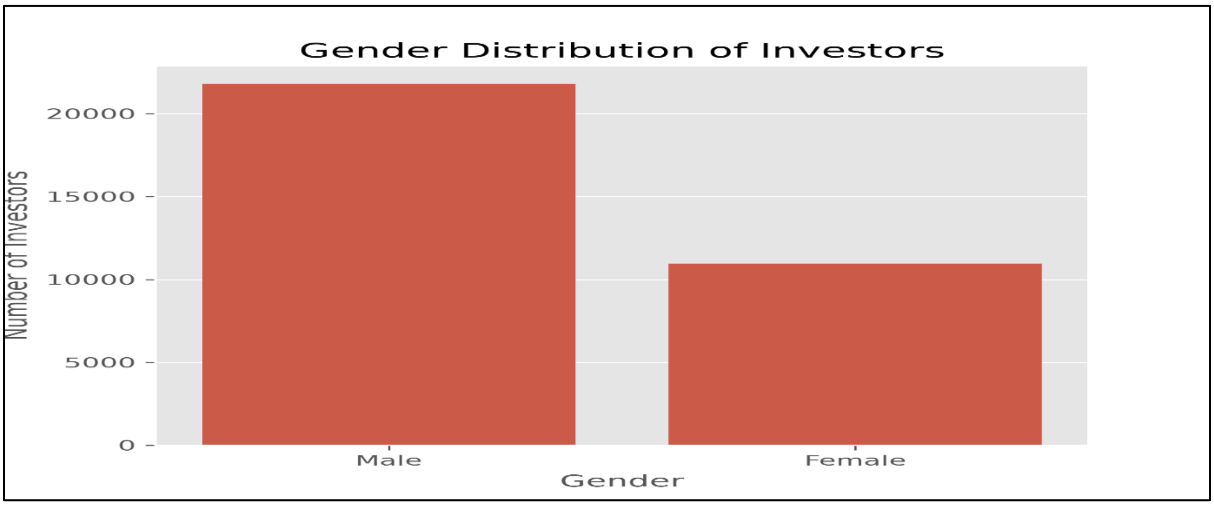
Monthly SIP inflows remained consistently strong throughout the observation period, reflecting increasing investor participation and the growing preference for disciplined long-term investing through Systematic Investment Plans.

Key Observations

- SBI Mutual Fund managed the highest Assets Under Management during the study period.
- Industry-wide AUM showed consistent growth over multiple years.
- Monthly SIP inflows remained stable with gradual upward movement.
- Investor participation continued to increase through systematic investment plans.

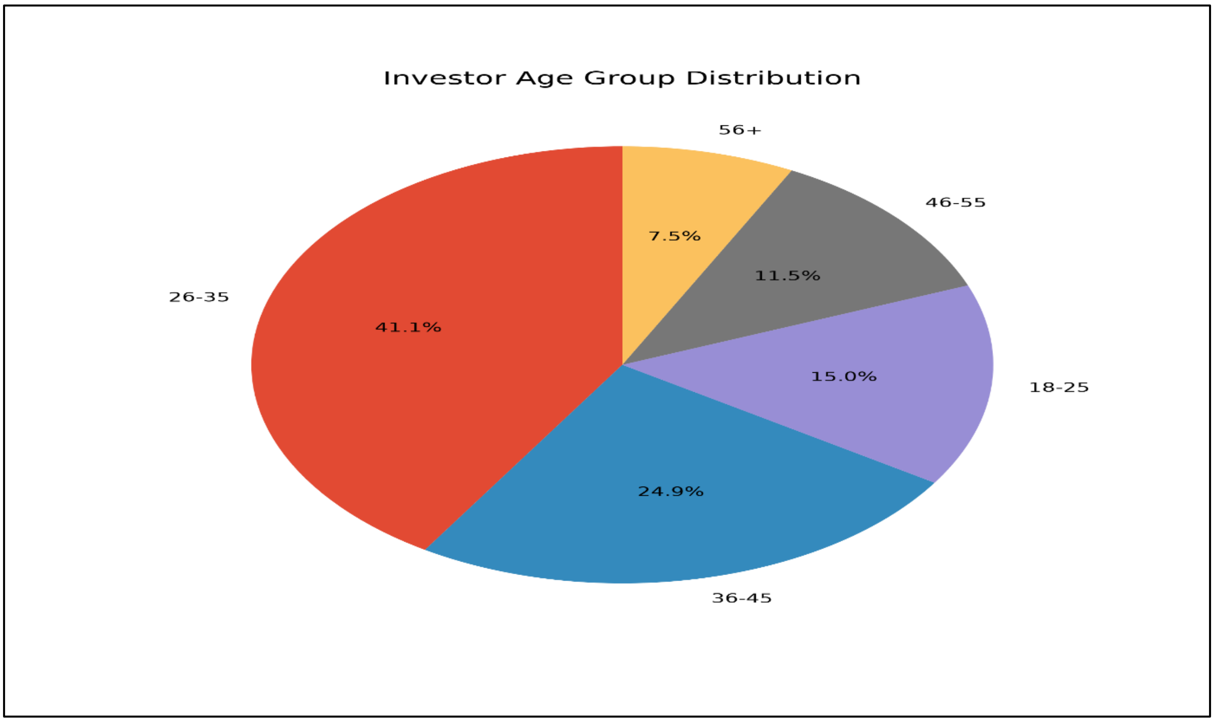
Investor Demographics Analysis

Figure 4.3 – Gender Distribution of Investors



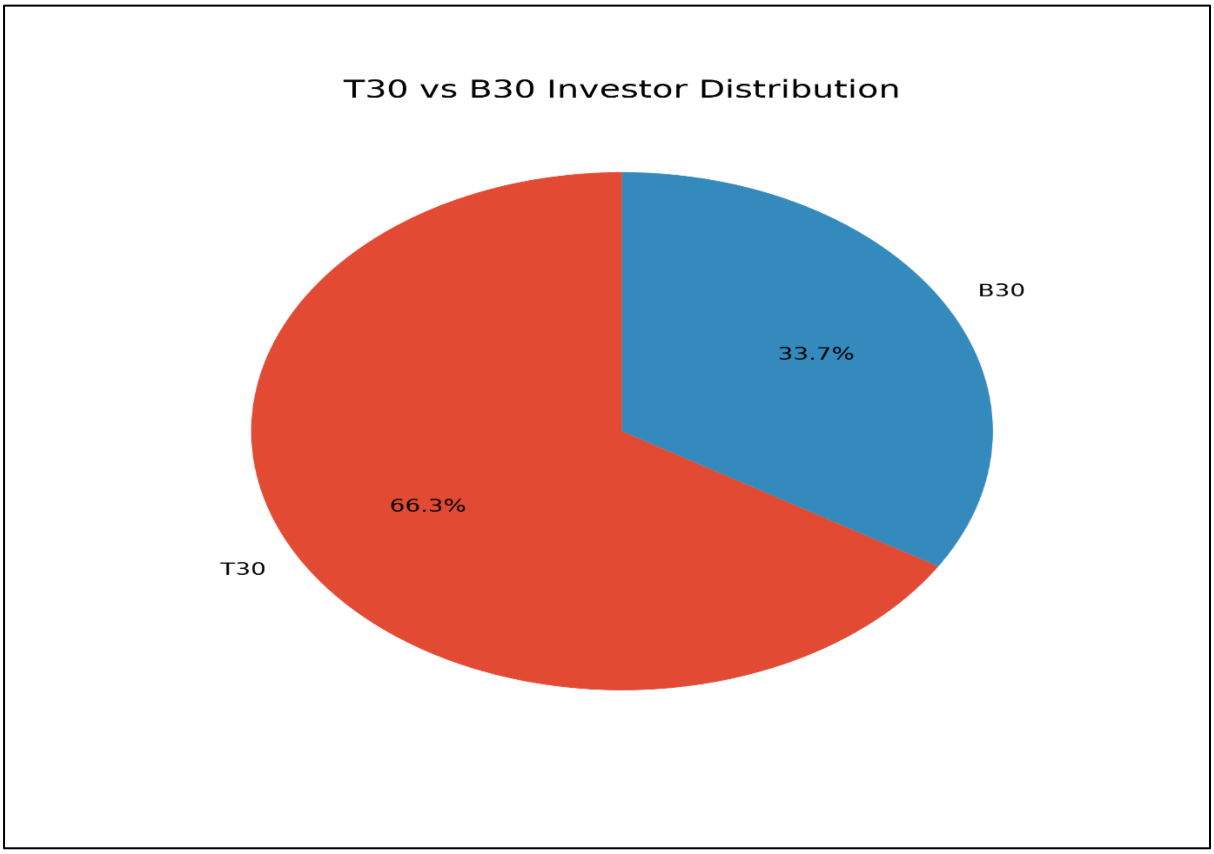
The gender distribution illustrates the participation of male and female investors in mutual fund investments, providing insights into investor diversity.

Figure 4.4 – Age Group Distribution



Most investments originate from economically active age groups, particularly investors between 26–35 years, indicating strong participation from working professionals.

Figure 4.5 – T30 vs B30 Distribution

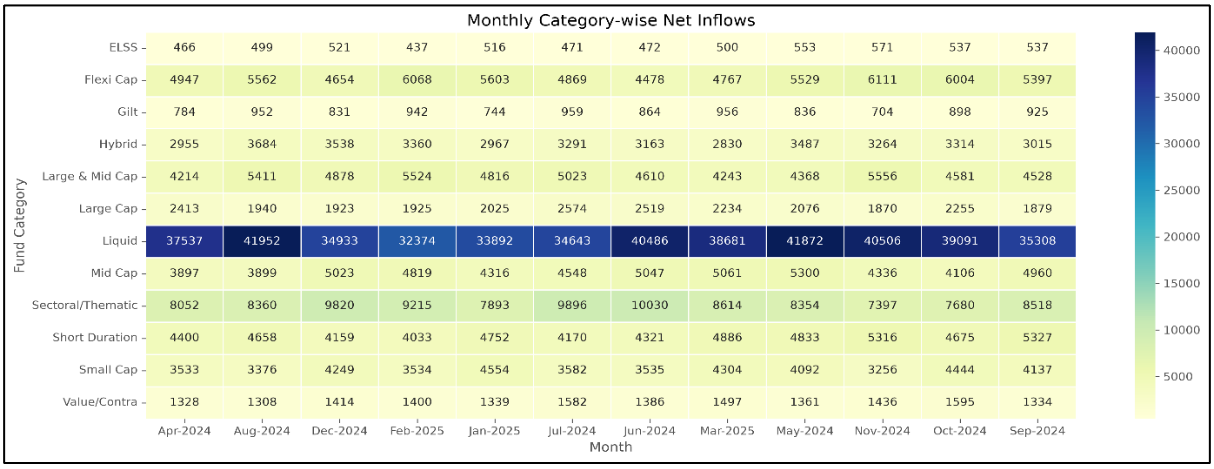


The comparison between T30 and B30 cities highlights investor participation across metropolitan and emerging markets, demonstrating expanding mutual fund penetration beyond major cities.

Key Observations

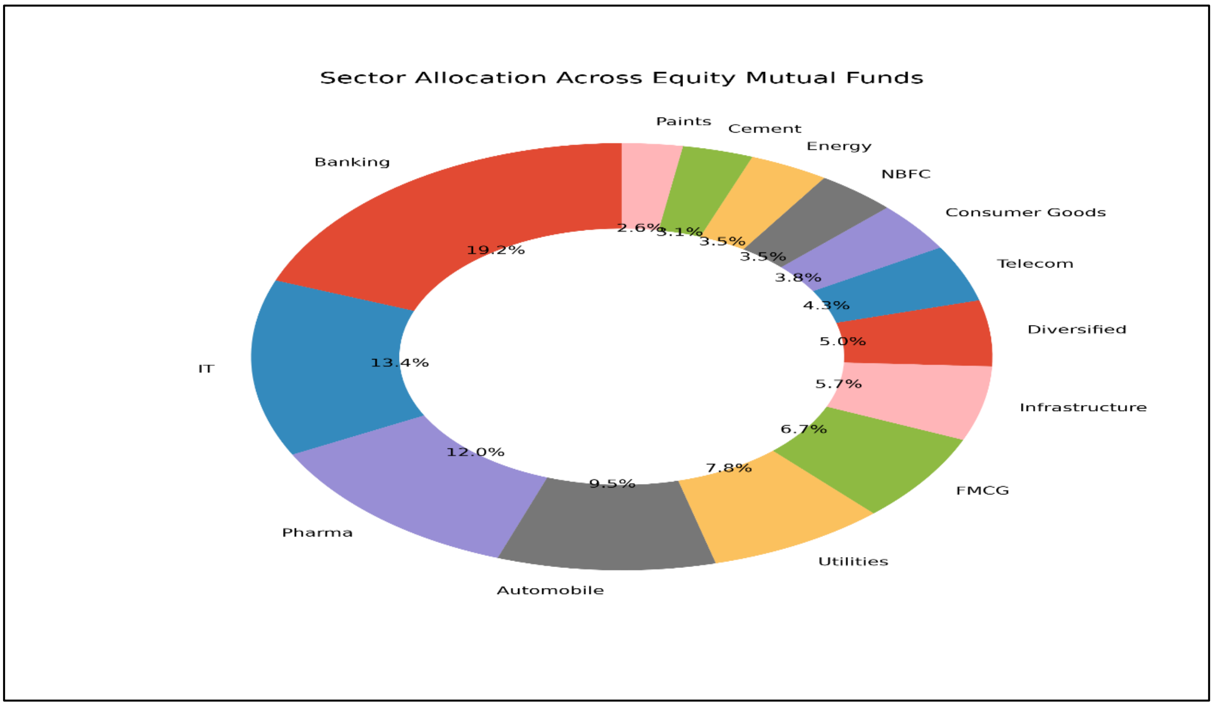
- Working-age investors represent the largest investment segment.
- Mutual fund participation is growing across both T30 and B30 cities.
- The investor base shows participation across multiple demographic groups.

Figure 4.6 – Category-wise Net Inflows



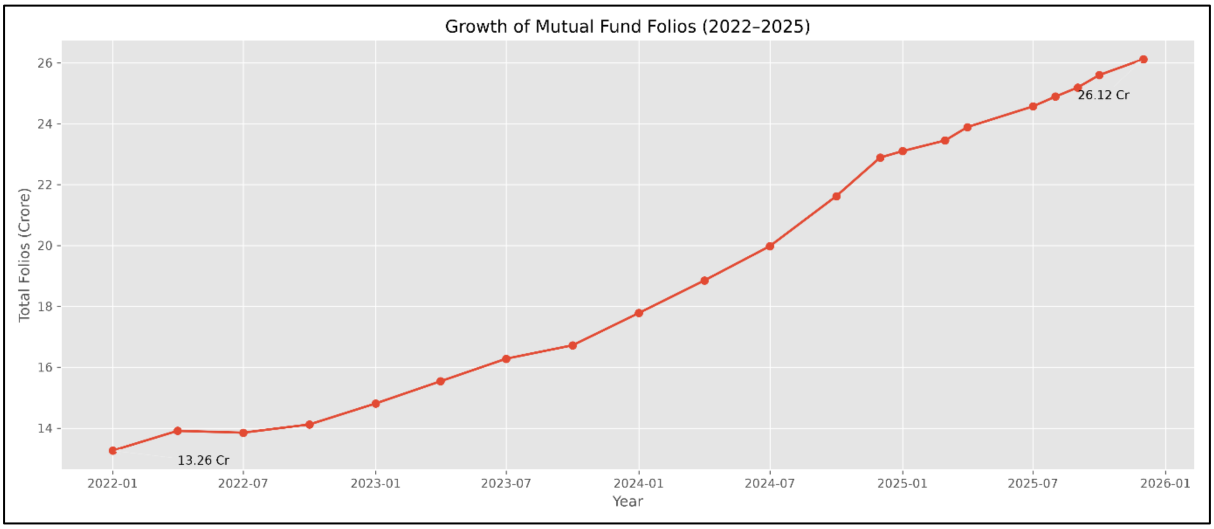
The heatmap illustrates monthly net inflows across different mutual fund categories. Flexi Cap, Mid Cap, and Large & Mid Cap funds consistently attracted higher investor inflows compared to other categories.

Figure 4.7 – Sector Allocation of Portfolio Holdings



The portfolio holdings are diversified across multiple sectors such as Banking, IT, Pharma, FMCG, Utilities, and Manufacturing. Sector allocation provides insight into diversification and investment concentration.

Figure 4.8 – Growth of Mutual Fund Folios (2022–2025)



Description:

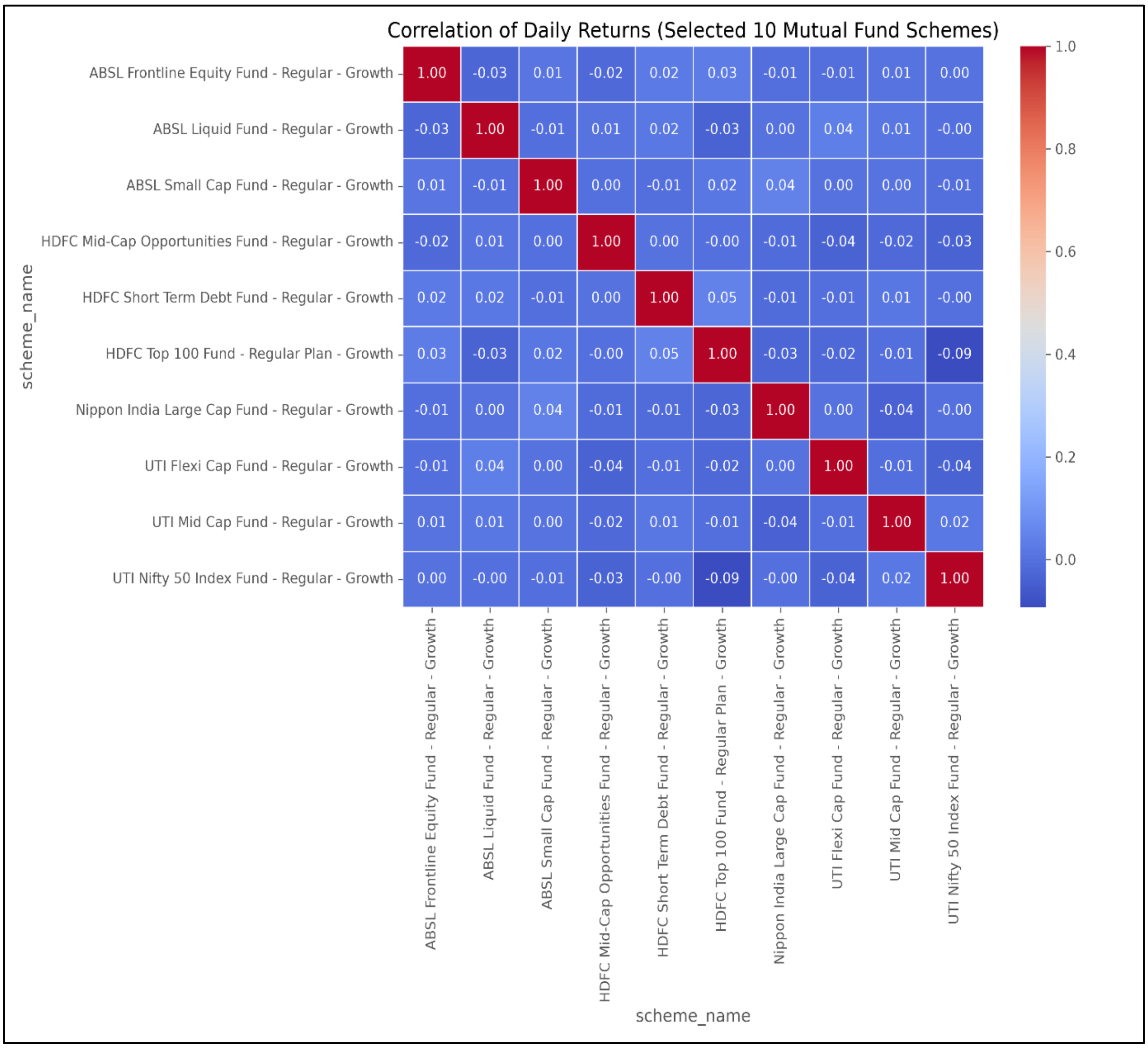
The chart illustrates the steady growth in the total number of mutual fund folios between 2022 and 2025. Total folios increased from approximately **13.26 crore** in early 2022 to over **26.12 crore** by the end of 2025, reflecting rising retail investor participation and increasing adoption of mutual fund investments in India.

Key Observations:

- Total mutual fund folios nearly doubled during the study period.
- Flexi Cap and Mid Cap funds consistently received strong net inflows.
- Banking and IT remain dominant sectors in equity portfolios
- The sharpest increase occurred throughout 2024 and 2025.
- Continuous folio growth indicates increasing investor confidence in mutual funds.
- Diversification across sectors reduces concentration risk
- Rising SIP participation is a major contributor to folio growth.

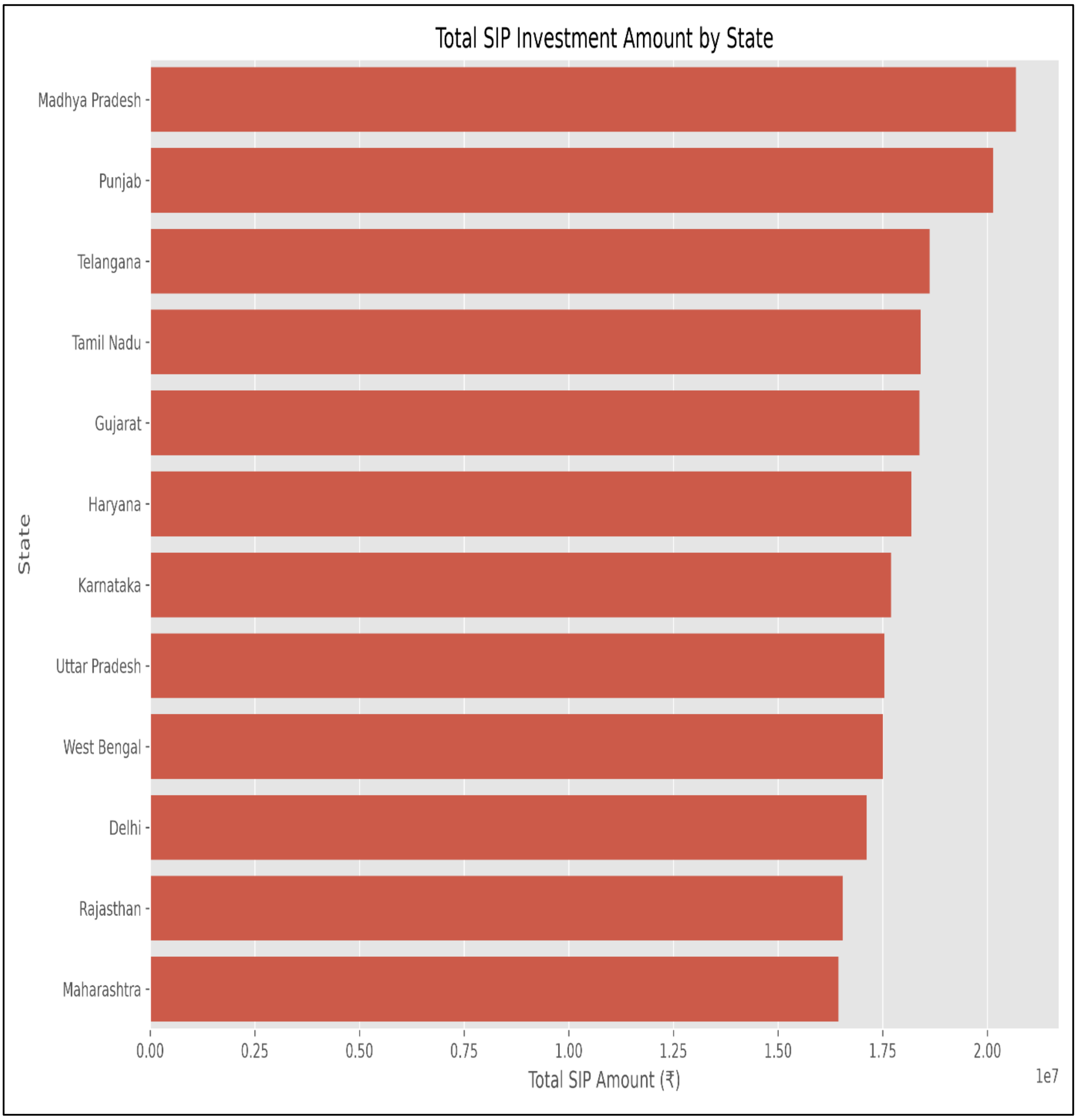
Investment Behaviour & Correlation Analysis

Figure 4.9 – NAV Return Correlation



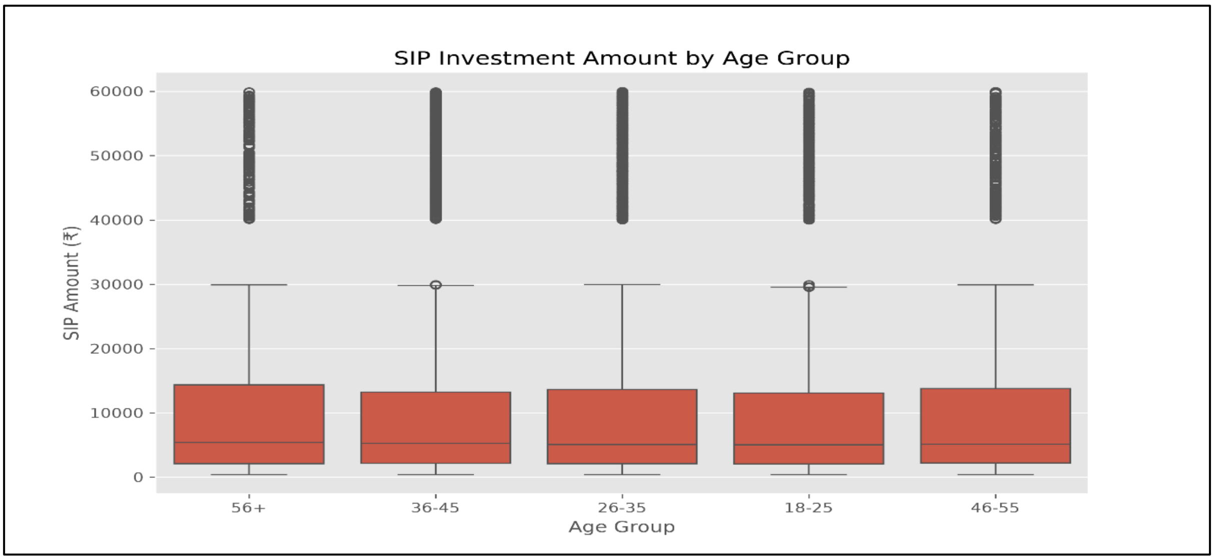
Correlation analysis measures the relationship between NAV movements and fund returns. Stronger positive correlations indicate consistent performance patterns across the selected schemes.

Figure 4.10 – State-wise SIP Distribution



The state-wise distribution highlights geographical participation in SIP investments. States with larger urban populations contributed a higher number of SIP transactions.

Figure 4.9 – SIP Amount Distribution by Age Group



The box plot compares SIP investment amounts across different age groups. Median SIP amounts are broadly similar across all age categories, while each group contains several high-value outlier investments. This indicates that investor age alone does not strongly determine SIP contribution size, although a small subset of investors in every age group invests significantly higher amounts.

Key Observations

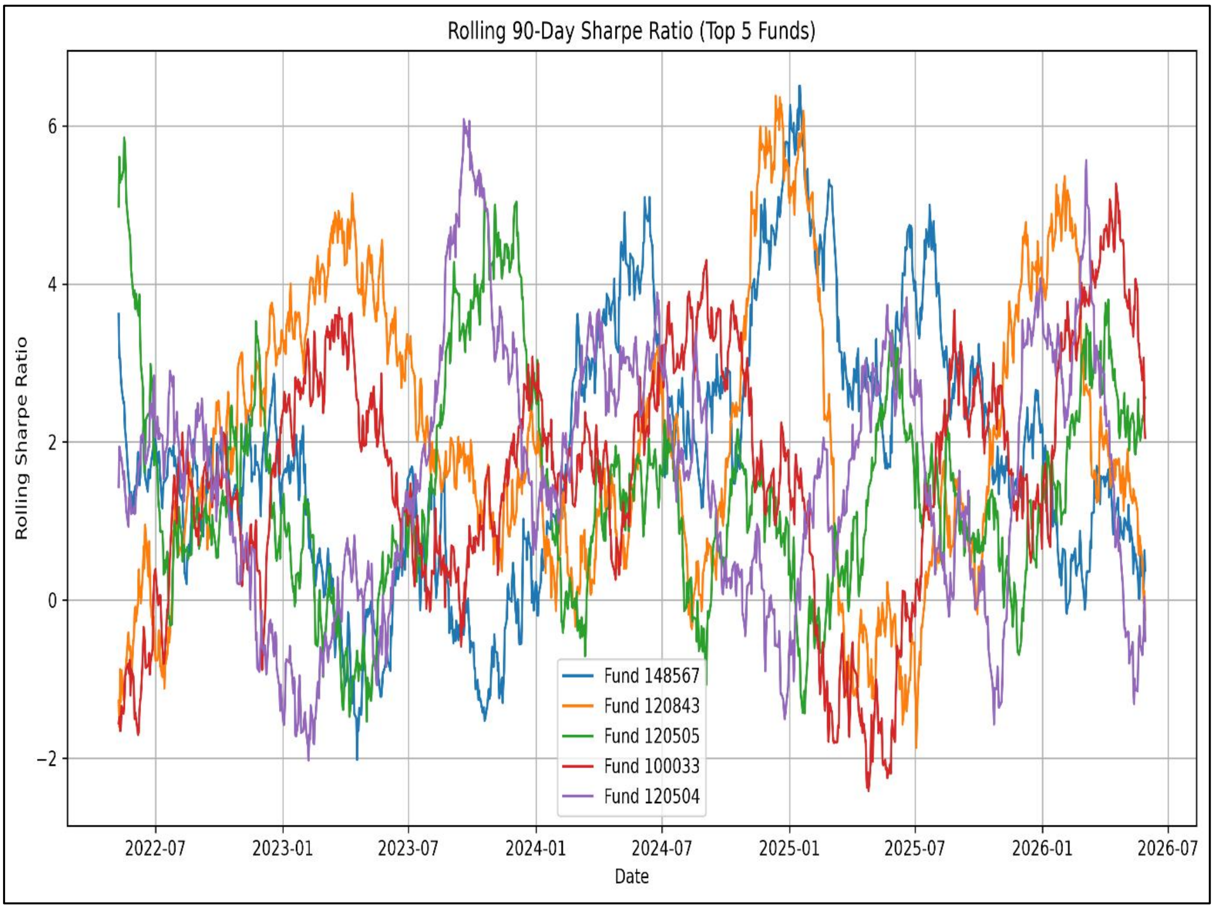
- NAV movements generally follow expected return behaviour.
- SIP participation is concentrated in financially active states.
- Most SIP investments fall within a moderate investment range, with relatively few large outliers.
- Investor behaviour indicates a preference for regular, long-term investing.

EDA Summary

The exploratory analysis confirmed that the datasets were clean, consistent, and suitable for financial analytics. The visualizations highlighted growing investor participation, increasing SIP adoption, diversified sector allocation, and strong market activity. These findings established a reliable foundation for the performance metrics and advanced risk analysis presented in the following sections.

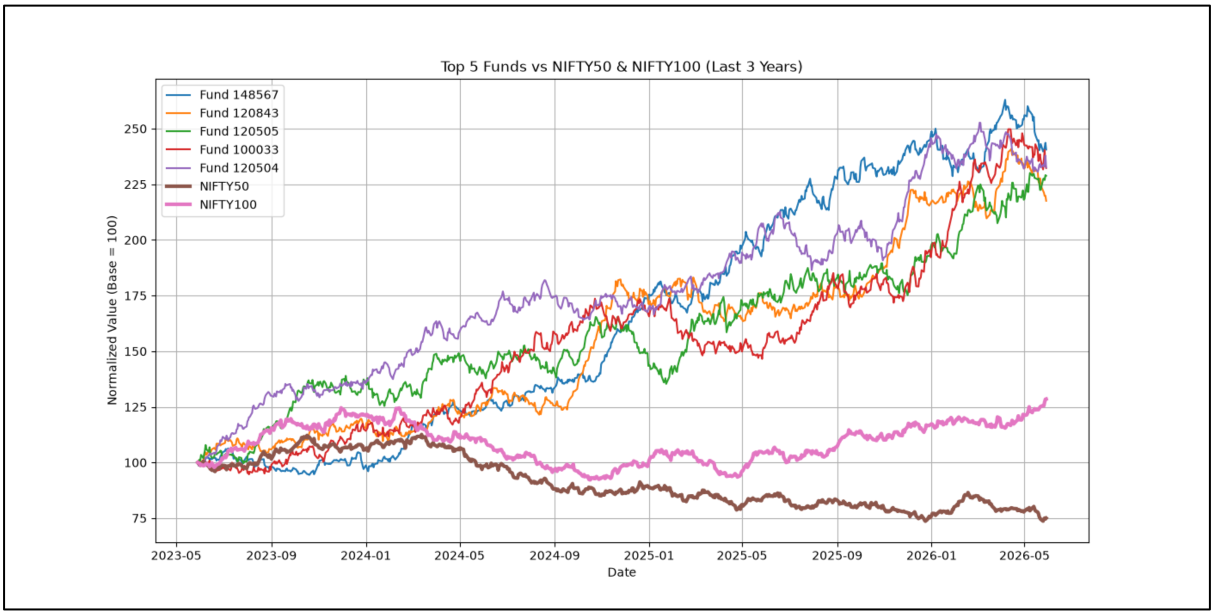
Performance Analytics

Figure 5.1 – Sharpe Ratio Comparison



The Sharpe Ratio measures the risk-adjusted return generated by each mutual fund. A higher Sharpe Ratio indicates that the fund delivers better returns for every unit of risk undertaken.

Figure 5.2 – Benchmark Performance Comparison



This comparison evaluates fund performance against the benchmark index. Funds consistently outperforming their benchmark demonstrate stronger portfolio management and investment efficiency.

Metric	**Source**
CAGR	cagr_comparison.csv
Sharpe Ratio	sharpe_ratio.csv
Alpha & Beta	alpha_beta.csv
Sortino Ratio	sortino_ratio.csv
Maximum Drawdown	maximum_drawdown.csv
Tracking Error	tracking_error.csv

Key Findings

- Risk-adjusted performance varies across different mutual fund schemes.
- Higher Sharpe Ratio funds provide better returns relative to the level of risk.
- Positive Alpha values indicate fund managers outperforming their benchmark.
- Lower Maximum Drawdown reflects stronger downside protection during market declines.
- Tracking Error helps evaluate how closely a fund follows its benchmark index.

Advanced Analytics

Figure 6.1 – Value at Risk (VaR) & Conditional VaR (CVaR)

	amfi_code	VaR_95	CVaR_95
22	119599	-0.0269	-0.0324
17	119095	-0.0262	-0.0317
4	101207	-0.0260	-0.0325
11	118634	-0.0254	-0.0323
21	119598	-0.0245	-0.0306
39	149324	-0.0235	-0.0310
7	102886	-0.0192	-0.0233
2	100033	-0.0190	-0.0235
25	120505	-0.0189	-0.0243
16	119094	-0.0185	-0.0243

Value at Risk (VaR) estimates the maximum expected portfolio loss at a given confidence level over a specified period. Conditional Value at Risk (CVaR) measures the average loss when losses exceed the VaR threshold, providing insight into extreme downside risk.

Figure 6.2 – Portfolio Concentration (HHI)

	amfi_code	HHI	scheme_name	category
11	119092	0.206448	Axis Bluechip Fund - Regular - Growth	Equity
3	101207	0.200700	ABSL Small Cap Fund - Regular - Growth	Equity
18	119599	0.174751	SBI Small Cap Fund - Direct Plan - Growth	Equity
4	102885	0.174709	UTI Nifty 50 Index Fund - Regular - Growth	Equity
7	118632	0.168298	Nippon India Large Cap Fund - Regular - Growth	Equity
29	148568	0.167930	Mirae Asset Emerging Bluechip Fund - Regular -...	Equity
21	120505	0.157570	ICICI Pru Midcap Fund - Regular - Growth	Equity
22	120506	0.153794	ICICI Pru Value Discovery Fund - Regular - Growth	Equity
27	125498	0.152414	HDFC Mid-Cap Opportunities Fund - Direct - Growth	Equity
23	120841	0.149680	Kotak Bluechip Fund - Regular - Growth	Equity

The Herfindahl–Hirschman Index (HHI) was used to measure the concentration of portfolio holdings. Lower HHI values indicate better diversification, whereas higher values indicate that a portfolio is concentrated in fewer securities and therefore carries relatively higher concentration risk

Figure 6.3 – Mutual Fund Recommendation System

	Scheme Name	3Y CAGR (%)	Sharpe Ratio	Alpha	Expense Ratio (%)	Fund Score
0	Mirae Asset Large Cap Fund - Regular - Growth	33.99	1.4483	0.2698	1.46	87.38
1	Kotak Flexicap Fund - Regular - Growth	29.58	1.3067	0.2733	1.45	83.12
2	ICICI Pru Midcap Fund - Regular - Growth	31.77	1.1801	0.2926	1.36	82.62
3	HDFC Mid-Cap Opportunities Fund - Regular - Gr...	32.43	1.0937	0.2720	1.38	81.50
4	ICICI Pru Bluechip Fund - Direct - Growth	32.48	1.0265	0.2119	0.80	80.38
5	Axis Midcap Fund - Regular - Growth	35.10	0.9982	0.2608	1.38	77.75
6	SBI Bluechip Fund - Regular Plan - Growth	30.45	1.2083	0.2320	1.54	77.25
7	Mirae Asset Tax Saver Fund - Regular - Growth	29.17	1.2349	0.2827	1.60	76.50
8	ABSL Frontline Equity Fund - Regular - Growth	28.96	1.0272	0.2140	1.60	71.00
9	DSP Midcap Fund - Regular - Growth	26.86	1.1321	0.2660	1.61	69.75

A rule-based recommendation engine was developed to rank mutual funds using multiple performance indicators including 3-Year CAGR, Sharpe Ratio, Alpha, Expense Ratio, and the overall Fund Score. This approach provides investors with a balanced recommendation by considering both returns and risk-adjusted performance rather than relying on a single metric.

Key Insights

- VaR and CVaR quantify the downside risk associated with mutual fund investments.
- HHI analysis indicates the level of diversification within each mutual fund portfolio.
- Funds with lower concentration generally provide better diversification benefits.
- The recommendation engine ranks funds using multiple quantitative performance indicators.
- The Fund Score combines return, risk-adjusted performance, expense ratio, and alpha into a single ranking metric.
- These advanced analytics assist investors in making informed and data-driven investment decisions.

Conclusion & Recommendations

This Mutual Fund Analytics Capstone Project successfully developed a complete data analytics pipeline for analysing mutual fund performance, investor behaviour, and market trends. The project integrated data ingestion, cleaning, storage, exploratory data analysis, performance evaluation, advanced risk analytics, and interactive dashboards into a single workflow.

Performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Maximum Drawdown, Tracking Error, Value at Risk (VaR), and Conditional Value at Risk (CVaR) were computed to evaluate mutual fund performance comprehensively. Investor cohort analysis, SIP continuity analysis, portfolio concentration analysis using the Herfindahl–Hirschman Index (HHI), and a rule-based recommendation system further enhanced the analytical capabilities of the project.

The Power BI dashboard enables interactive exploration of fund performance, investor demographics, portfolio diversification, and SIP market trends. Overall, the project demonstrates practical applications of Python, SQL, data visualization, and business intelligence techniques in the financial services domain.

Key Project Outcomes

- Successfully built an automated ETL pipeline.
- Loaded cleaned datasets into a SQLite database.
- Performed comprehensive Exploratory Data Analysis (EDA).
- Calculated major mutual fund performance metrics.
- Built an interactive Power BI dashboard with multiple pages.
- Implemented advanced analytics including VaR, CVaR, HHI, cohort analysis, SIP continuity analysis, and a recommendation engine.
- Generated reports and visualisations for investment decision support.

Business Recommendations

1. Investors should evaluate risk-adjusted returns rather than relying only on historical returns.
2. Diversified portfolios with lower concentration (HHI) are generally less risky.
3. Regular SIP investments help reduce market timing risk.
4. Investors should consider expense ratios while selecting mutual funds.
5. Periodic portfolio reviews are recommended to maintain diversification and risk alignment.
6. Financial institutions can leverage predictive analytics to personalise fund recommendations.

Future Scope

The project can be extended by integrating live AMFI APIs for real-time NAV updates, implementing machine learning models for return prediction, performing portfolio optimisation using the Markowitz Efficient Frontier, conducting Monte Carlo simulations for future NAV forecasting, and developing a Streamlit-based web application for interactive portfolio analysis.

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